

River Tyne Plastic Pellet (Nurdle) Spill – Scientific Briefing for Volunteers

Tyneside Coast (Longsands • Tynemouth • Cullercoats • Whitley Bay • South Shields • Sunderland)

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What happened?

On 19 July 2026, the offshore support vessel Wind Orca collided with the berthed container ship BG Orange inside the Port of Tyne.

The collision damaged a dockside crane and ruptured a container carrying industrial plastic resin pellets ("nurdles").

An estimated 24–25 tonnes, equivalent to roughly one billion pellets, escaped into the River Tyne before containment was possible. **[1, 2]**

What are nurdles?

Nurdles are tiny plastic resin pellets measuring approximately 1–5 mm.

They are the raw material used to manufacture almost every plastic product, including:

Bottles, food containers, packaging, toys, pipes, automotive parts, medical plastics, etc.

Therefore, they are not waste plastic—they are virgin industrial feedstock.

Why have they reached so many beaches?

Once the pellets entered the Tyne Estuary they behaved exactly as marine scientists would predict. They were transported by:

Tidal currents, wave action, prevailing winds and longshore coastal drift

Because they are extremely small and buoyant, they accumulate naturally at:

The high tide line, seaweed deposits, shell accumulations, rocky crevices and dunes.

This explains why pellets have been reported at:

South Shields, Tynemouth Longsands, King Edward's Bay, Cullercoats, Whitley Bay, Seaburn, Sunderland and may continue spreading further along the North Sea coast. **[1, 3]**

Are they toxic?

To humans:

Current official advice indicates the pellets themselves present a very low direct health risk. Touching them occasionally is not expected to cause poisoning.

Nevertheless:

- Avoid prolonged skin contact.
- Wash hands afterwards.
- Keep children from playing with them.
- Prevent pets from eating them.

Environmental risk:

The environmental concern is high.

Nurdles: absorb oil, absorb diesel residues, absorb persistent organic pollutants, adsorb heavy organic contaminants

Marine organisms mistake them for food.

Affected wildlife may include: seabirds, fish, crabs, shellfish and marine mammals indirectly through the food chain. [1]

Potential impacts include:

Digestive blockage, starvation, reduced reproduction, transfer of pollutants through the food web and fragmentation into microplastics.

These long-term ecological effects are why nurdle spills are treated as major environmental incidents.

Why is recovery so difficult?

Unlike an oil spill: they do not evaporate or dissolve, and they float, sink, and resurface repeatedly. Storms, on the other hand, can rebury and redistribute them.

Even an excellent clean-up cannot recover every pellet.

Experience worldwide shows that pellets may remain in coastal sediments for many years.

What are the official organisations doing?

Several organisations are involved:

Port of Tyne:

Activated its spill response plan.

Deployed boats to recover pellets from the river.

Working with partner agencies to contain further spread. [2]

Environment Agency:

Providing technical advice.

Supporting local authorities during clean-up.

Participating in the investigation. [1, 4]

HM Coastguard:

Coordinating pollution response with local authorities along affected beaches. [1]

South Tyneside Council:

Assessing contamination.

Organising appropriate clean-up.

Intends to recover clean-up costs from those responsible under the "polluter pays" principle. [1, 4]

North Tyneside Council:

Monitoring affected beaches.

Coordinating local responses where pellets have arrived. [5]

Why were volunteers initially advised not to collect them?

Early in the response, authorities needed to:

Assess the extent of contamination.

Preserve evidence for the investigation.

Avoid spreading pellets inadvertently.

Prevent damage to sensitive habitats.

Establish safe disposal procedures.

- As organised clean-up plans developed, volunteer participation became more structured in many areas.

Best practice for volunteers:

If participating in an organised clean-up:

- ✓ Wear gloves.
- ✓ Wear sturdy footwear.
- ✓ Keep pellets separate from normal litter.
- ✓ Use buckets, dustpans, soft brushes or fine mesh scoops where appropriate.
- ✓ Avoid disturbing sand dunes and bird nesting areas.
- ✓ Keep dogs on leads.
- ✓ Wash hands thoroughly afterwards.
- ✓ **Follow instructions from local organisers and councils.** (And Join Local WhatsApp Groups)

Avoid using powerful rakes or digging deeply into dunes, as this can damage fragile coastal habitats.

Why is this considered a serious environmental incident?

Although each pellet is tiny, the scale is enormous:

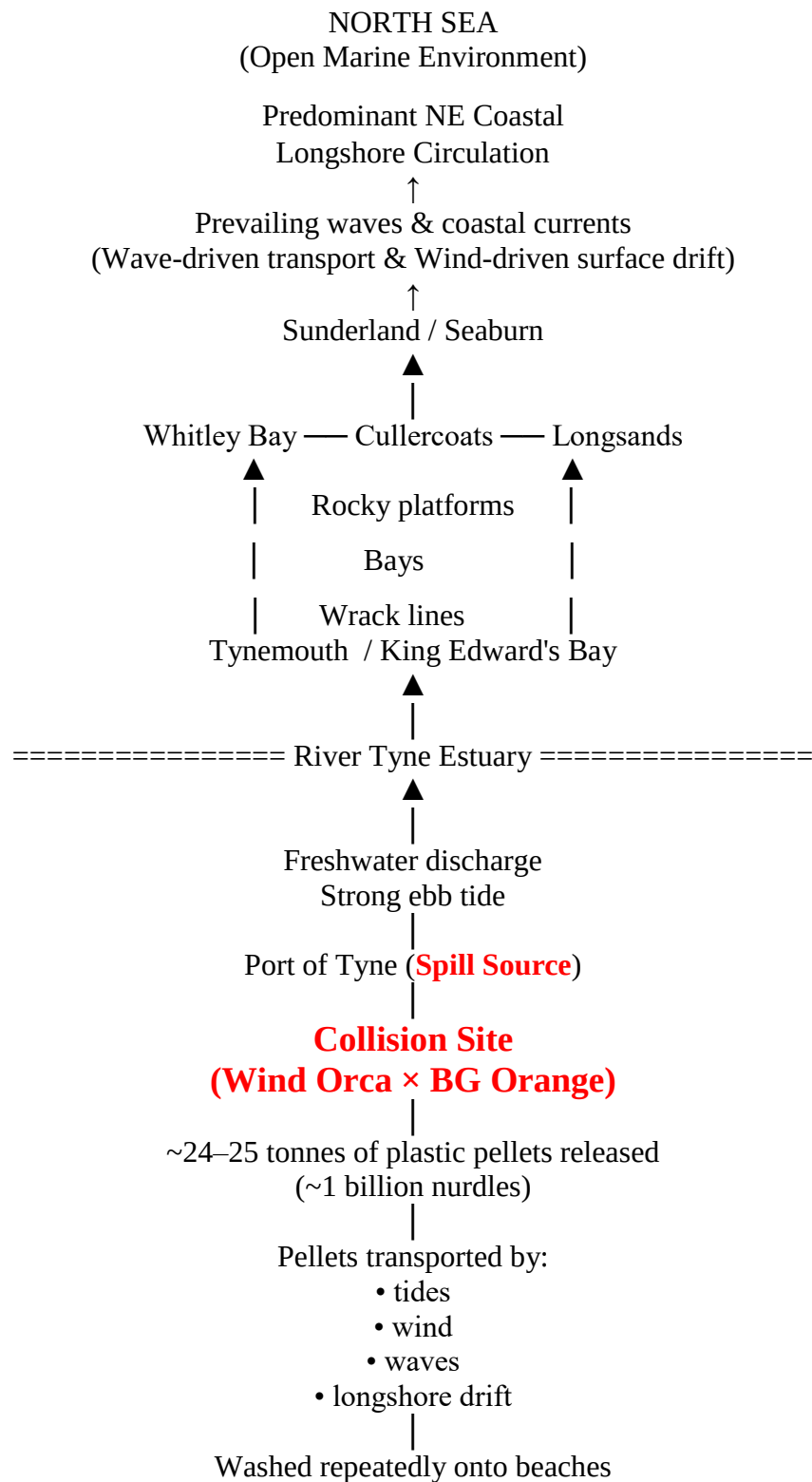
Approximately 1,000,000,000 pellets. Roughly 25 tonnes.

Capable of spreading over many kilometres of coastline.

Extremely difficult to recover completely.

Marine scientists regard nurdle spills as one of the most persistent forms of plastic pollution because the pellets continue circulating between beaches and the sea for years.

Below is the **Scientific Map showing why Longsands, Cullercoats, Whitley Bay and nearby beaches are among the first locations expected to accumulate nurdles**



Key message:

This incident is not simply a litter problem. It is a **significant marine pollution** event resulting from a shipping accident. Every properly collected bag of pellets helps reduce the amount available to enter the food chain or break down into microplastics. At the same time, **clean-up efforts should be coordinated with local authorities to maximise effectiveness and minimise disturbance to sensitive coastal habitats.** [1, 2, 5]

References:

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